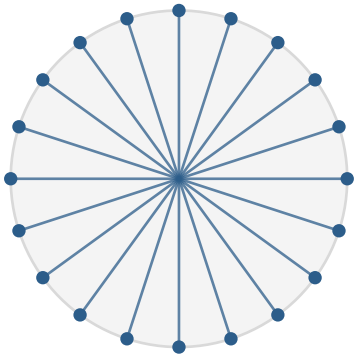
RT three-fold axes: an open A2 / su(3) candidate

1. RT axis inventory



10 three-fold axes
20 vertices as opposite pairs

2. Seeds, Gram rank, closure

Gram rank(G) = 6 (Sym² bound)

↓

traceless seed span 5
+ 3 commutator generators

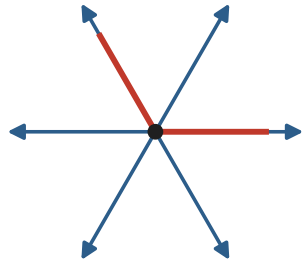
↓

closure dim 5 + 3 = 8

Closure dim 8 is not the Lie rank of su(3).

3. A2 fingerprint

Cartan matrix = [[2, -1], [-1, 2]]



6 roots (3 positive + 3 negative)
2 Cartan generators

RT witness	Controls	Matched 10-axis control	Proof status
10 axes, closure 8D **A2 match**	**Cube / octahedron** **/ cuboctahedron** **not RT-specific**	**200 / 200 reach 8D + A2** **not RT-specific**	**basis dim 8; Jacobi pass** **O.39 open**

The closure and A2 fingerprint are not specific to the RT axes; the color identification is an open Tier 3 candidate (O.39).

Sources: Ch03 Color Force SU(3); protocol_gauge_uniqueness_results.json; protocol_su3_proof_complete_results.json.